COMPUTER NETWORKS
Common to CSE&IT

V Semester | | **L** | **T** | **P** | **C**
Course Code: 201CS5T01 | | 3 | 0 | 0 | 3

Course Outcomes:

At the end of the Course, Student will be able to:

CO1: Describe network topologies, reference models and media for data transmission.
CO2: Analyze error and flow control issues in data link layer.
CO3: Classify MAC protocols and wired LAN technologies based on performance.
CO4: Apply routing algorithms and congestion control techniques for effective data transmission.
CO5: Analyze protocols used in the layers of reference models.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-
CO4	3	2	3	1	-	-	-	-	-	-	-	-
CO5	2	2	-	-	-	-	-	-	-	-	-	-

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	1	-
CO2	2	-
CO3	1	-
CO4	2	-
CO5	2	-

Unit - I
Introduction: Network Types, LAN, MAN, WAN, Network Topologies Reference models- The OSI Reference Model, the TCP/IP Reference Model , A Comparison of the OSI and TCP/IP Reference Models, OSI Vs TCP/IP, Lack of OSI models success, Internet History.
Physical Layer: Introduction to Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and unguided media: Wireless-Radio waves, microwaves, infrared.

Unit – II
Data link layer: Design issues, Framing: fixed size framing, variable size framing, flow control, error control, error detection and correction codes, CRC, Checksum: idea, one's complement internet checksum, services provided to Network Layer, Elementary Data Link Layer protocols: simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel.Sliding window protocol: One bit, Go back N, Selective repeat-Stop and wait protocol,
Data link layer in HDLC: configuration and transfer modes, frames, control field, point to point protocol (PPP): framing transition phase, multiplexing, multi link PPP.

Unit – III
Media Access Control: Random Access: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance,
Controlled Access: Reservation, Polling, Token Passing, Channelization: frequency division multiple Access(FDMA), time division multiple access(TDMA), code division multiple access(CDMA).
Wired LANs: Ethernet, Ethernet Protocol, Standard Ethernet, Fast Ethernet(100 Mbps), Gigabit Ethernet, 10 Gigabit Ethernet.

Unit – IV

The Network Layer Design Issues: Store and Forward Packet Switching, Services Provided to the Transport layer, Implementation of Connectionless Service, Implementation of Connection Oriented Service, Comparison of Virtual Circuit and Datagram Networks, Routing Algorithms: The Optimality principle, Shortest path, Flooding, Distance vector, Link state, Hierarchical, Congestion Control algorithms: General principles of congestion control, Congestion prevention polices, Approaches to Congestion Control, Traffic Aware Routing, Admission Control, Traffic Throttling, Load Shedding, Traffic Control Algorithm:Leaky bucket & Token bucket.

Internet Working: Network layer in the internet, IP protocols: IP Version 4,IP Version 6,Transition from IPV4 to IPV6, Comparison of IPV4 & IPV6,Internet control protocols: ICMP, ARP, DHCP

Unit – V

The Transport Layer: Transport layer protocols: Introduction, services, port number, User data gram protocol: UDP services, UDP applications,

Transmission control protocol: TCP services, TCP features, Segment, A TCP connection, windows in TCP, flow control, Error control, Congestion control in TCP.

Application Layer: World Wide Web: HTTP, Electronic mail, Architecture, web based mail, email security, TELENET, local versus remote Logging, Domain Name System: Name Space, DNS, SNMP.

Text Books:

1. Computer Networks — Andrew S Tanenbaum and David J Wetherall, 5th Edition, Pearson Education, 2013.
2. Data Communications and Networking – Behrouz A.Forouzan, 5th Edition, McGraw Hill Education, 2012.

Reference Books:

1. Data Communications and Networks- Achut S Godbole, AtulKahate
2. Computer Networks, Mayank Dave, CENGAGE
3. An Engineering Approach to Computer Networks-S. Keshav, 2nd Edition, Pearson Education.

Web Links:

1. https://nptel.ac.in/courses/106105081
2. https://www.coursera.org/learn/fundamentals-network-communications
3. https://nptel.ac.in/courses/106/106/106106091/
4. https://www.udemy.com/course/mta-networking-fundamentals/